STUDY · ENGINE/STUDIES/ONE-CUP · BRANCH CCR-82C69111-N0P4Q2

One cup, five worlds, one frame

What the film is, how each frame is made, why it takes the time it takes, and what we fixed on the way. The film is ten seconds long. The cup stays still while the world around it changes on hard cuts.



Preview film (360p, 6 samples): rendering now, ~10 min

Final film: **~1 hour** after the preview

Old Blender room film: **~10 hours** for 20 s

THE CUT · 10.0 S AT 24 FPS · 240 FRAMES

The match cuts

Same lens, same frame, one slow push-in across every cut. Each cut lands on a real recording of a cup being set down. The cuts get shorter (3.0 s, 2.0 s, 1.4 s, 1.1 s), then night holds and fades out.



00.0 – 03.0

Dawn

Low sun through a blind, black void. Full cup, strong steam. Birds.



03.0 – 05.0

Rain

Overcast window, rain on the glass. Coffee 68%. Rain heard from inside.



05.0 – 06.4

The room

The tutorial desk: blinds, monitor. 50%. Room tone.



06.4 – 07.5

Golden

Sun behind, rim light, dust. 32%. Evening birds.



07.5 – 10.0

Night

Desk lamp, city out of focus. 12%, no steam. Traffic. Fades out.

0 s 2 4 6 8 10 s

STORY

What happens, and what is missing

Right now the story is "one cup, one whole day": the light changes, the coffee drops and the steam fades. That is a mood, not an event. You asked for something to happen. Here are three small actions that fit this frame and stay cheap to build. Each one is a single thing that moves.

RECOMMENDED

A hand takes a sip on every cut

A hand lifts the cup out of frame, and the cut happens while it is gone. The cup comes back down into a new world on the cup sound. The level is lower each time. The action and the cut become the same gesture.

OPTION

A spoon stirs, and the swirl carries the cut

A spoon turns once in the coffee. The swirl continues across the cut into the next world, and the stir slows down as the day goes on.

OPTION

One pour at dawn, then the day

A kettle pours into the empty cup in the first shot, then the cuts start. This is the most visual option but the most work, because the stream of coffee is a liquid effect.

PROCEDURE

How one frame is made

The slow part, soft light bouncing around the room, is calculated once per world and stored on the model. Every frame then only pays for what moves or must stay sharp. This is the idea we took from room-at-dawn.

ONCE	ONCE PER WORLD	EVERY FRAME	EVERY FRAME	ONCE
Build	**Bake the soft light**	**24 sub-frames**	**Grade**	**Finish**
Each world is a data file: cup, table, walls, window, blinds, lamp, and how full the coffee is.	A CPU path tracer follows light from the window, lamp and walls, and stores it on every point of the model (about 110k each).	The frame is drawn 24 times, each a little different (see below), and averaged together.	Halation (a thin red bleed at bright edges), glow, a filmic curve that rolls highlights off, vignette, grain.	Encode at 24 fps. A recorded ambience for each world, cut with the picture, plus a cup-down sound on every cut.
`worlds.js`	`1–5 min per world`	`~0.8 s each, 720p`	`a few ms`	`~1 min`

SAMPLING

What "24 samples" means here

Each sample is a whole picture of the scene, drawn with five things moved slightly. When they are averaged, blur and softness come out of the real optics. No denoiser smears the image afterwards.

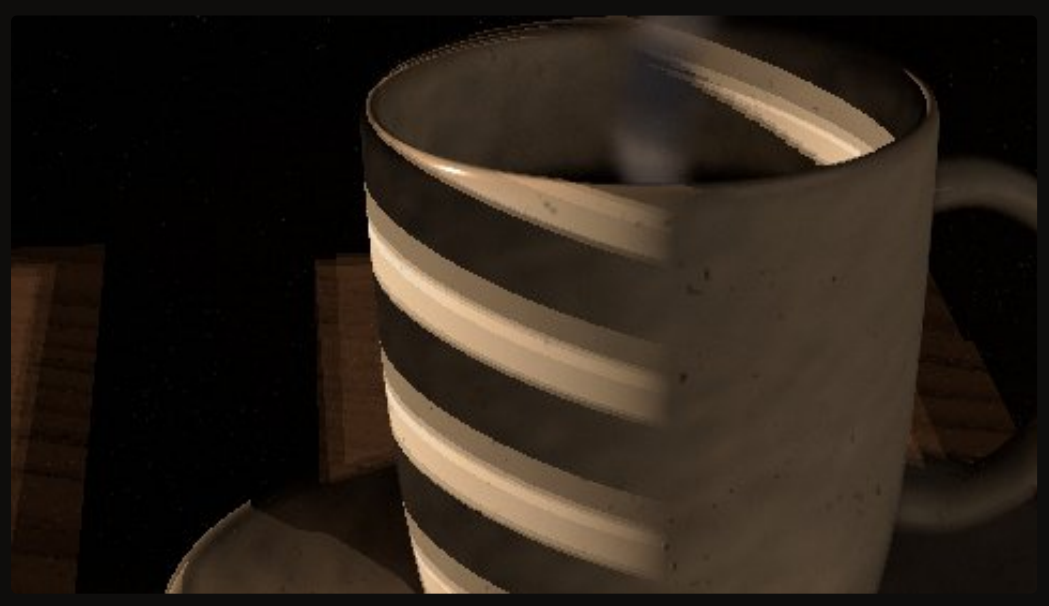
Pixel
Moved by under a pixel, for clean edges

Lens
The eye moves across an f/2 aperture, giving real depth of field and round bokeh

Shutter
Time spread over half a frame (180°), giving real motion blur

Sun disc
The sun moves inside its disc, so shadow edges soften naturally

Gloss and steam
A new random sample each pass, so reflections and steam converge



4 sub-frames. Grainy table sheen, stepped shadow edges.

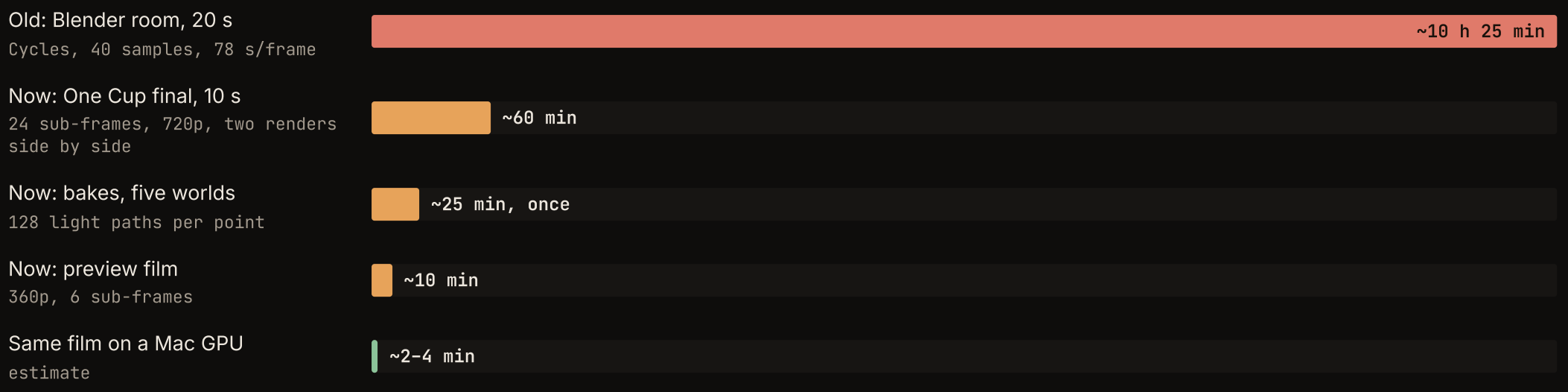


48 sub-frames. The same frame, converged. 24 is the middle ground for the film.

TIME · SAME SCALE FOR EVERY BAR

Where the hours go

This machine has no graphics card, so a software GPU on four CPU cores emulates WebGL. That is the whole reason for the waiting. The same code on a Mac with a GPU draws a sub-frame in milliseconds.



The bars share one scale: the full width is 10 hours 25 minutes.

SETTINGS

The numbers

SETTING	VALUE	WHY
Resolution	1280×720 render, 1080p delivery	Review size; the 1080p file is upscaled
Sub-frames per frame	24 (preview 6)	Smooth depth of field, shadows and gloss at a cost we can afford
Bake	128 paths/point, 4 bounces	Soft light from window, lamp, walls; a light denoise between neighbouring points
Lens	50 mm, f/2.0, focus on the cup front	Background falls soft; city lights become discs
Shutter	180°, 24 fps	Film motion blur; never crosses a cut
Sun	disc 0.0055 rad, live shadow map 2048²	Sharp blind stripes with real soft edges
Steam	30 march steps per pixel	Thin curling sheets, cut by the blind stripes
Grade	Uchimura curve, halation 0.55, glow 0.07	Highlights roll to white; bright rims bleed warm

FIXES

What went wrong, and how we found it

The cup rendered black

The triangles faced inward, so the renderer hid the outside of the cup and showed the dark inside of the back wall. Now every part's faces are checked against its normals before the first render.

BEFORE



AFTER



Bands across the table

The very low sun grazed the table and its shadow test misfired in long stripes. A slope-aware shadow tolerance fixed it. We found it by viewing the light alone on flat grey.

LIGHT-ONLY VIEW, BEFORE



AFTER



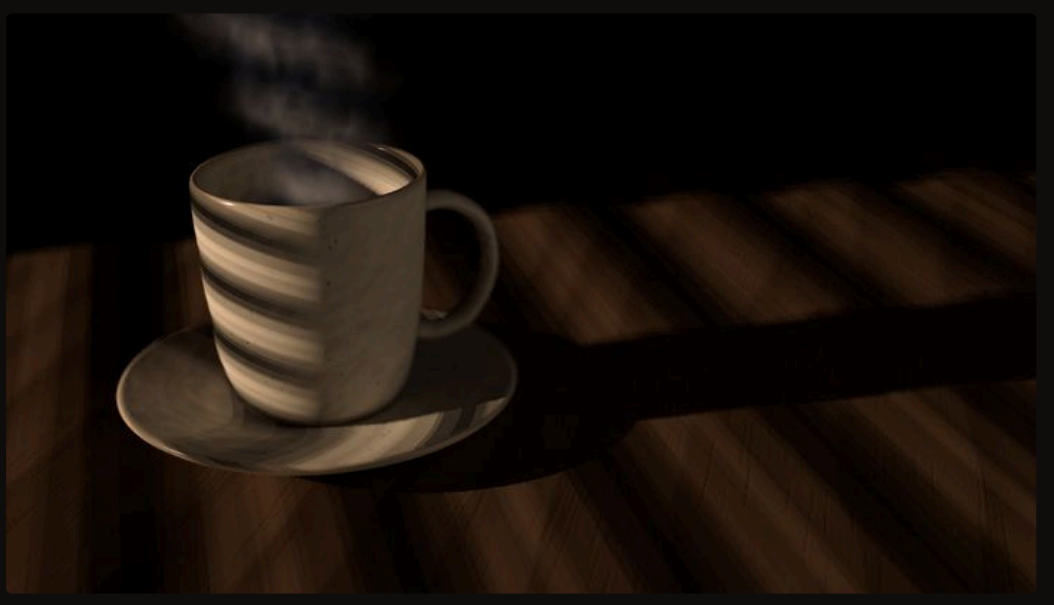
The table looked like plaid

The wood grain crossed the blind stripes. Turning the grain to run with the stripes fixed it.

BEFORE



AFTER



Night was spotty

Light bouncing inside the lampshade, a few centimetres from the bulb, produced rare, huge values that showed up as blotches. Capping those bounces fixed it. Before that, we found a worse bug: every world had been baked with dawn's light, because the bake's helper threads never received the world's name.

BEFORE



AFTER



LESSONS

What room-at-dawn taught us, and what we did wrong before

BEFORE (BLENDER ROOM)

- Recalculated all the light for every frame: 78 s each, 10 hours a film
- Low samples plus an AI denoiser gave a smeary, plastic look
- Bright windows clipped and washed out
- One long walk-through instead of designed shots
- Layout checked after rendering, so positions looked guessed

LEARNED FROM ROOM-AT-DAWN

- Calculate the slow, soft light once and store it
- Get blur and softness by averaging honest samples, never by filtering
- A filmic curve that rolls highlights off to white
- The film as data: shots, cameras and light in lists, and everything that moves as a function of time
- Every light has a real source in the scene

OURS, ON TOP

- Live sun and lamp, so stripes stay razor-sharp and can sway
- Exact window and blind reflections on every surface
- Lens-true bokeh, film halation, steam lit by the blind
- Five worlds from one engine, for match cuts
- Debug views (light only, steam only) before judging materials

Files: engine/studies/one-cup (worlds.js, scene.js) · tools/bake_one_cup.mjs · tools/one_cup_finish.sh · docs/STUDIES.md · sound credits in deliver/studies/SOUND_CREDITS.md